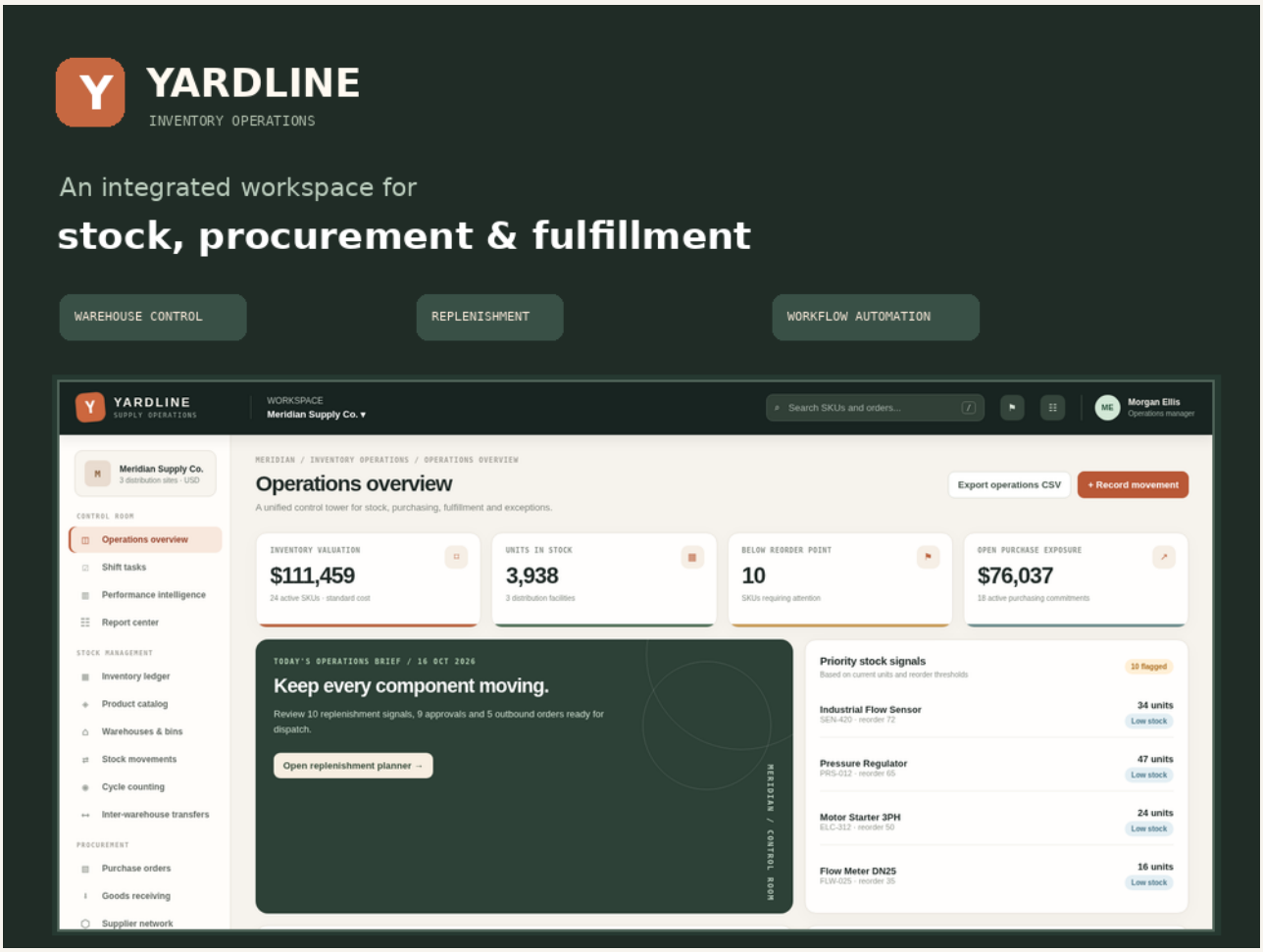
PROJECT 06 / PRODUCT CASE STUDY

Yardline

ERP & Inventory Workflow Automation

A connected inventory and purchasing control tower with 29 navigation modules, three warehouse locations, human approval points and integrated operational reporting.



PROJECT SCOPE

Inventory • Procurement • Fulfillment • Replenishment • Automation

01 / EXECUTIVE SUMMARY

The operating challenge

Industrial supply teams often manage stock levels, purchase orders, warehouse transfers, dispatches, supplier lead times and cycle counts in disconnected spreadsheets. Yardline consolidates those workflows into one inventory operations interface.

Solution overview

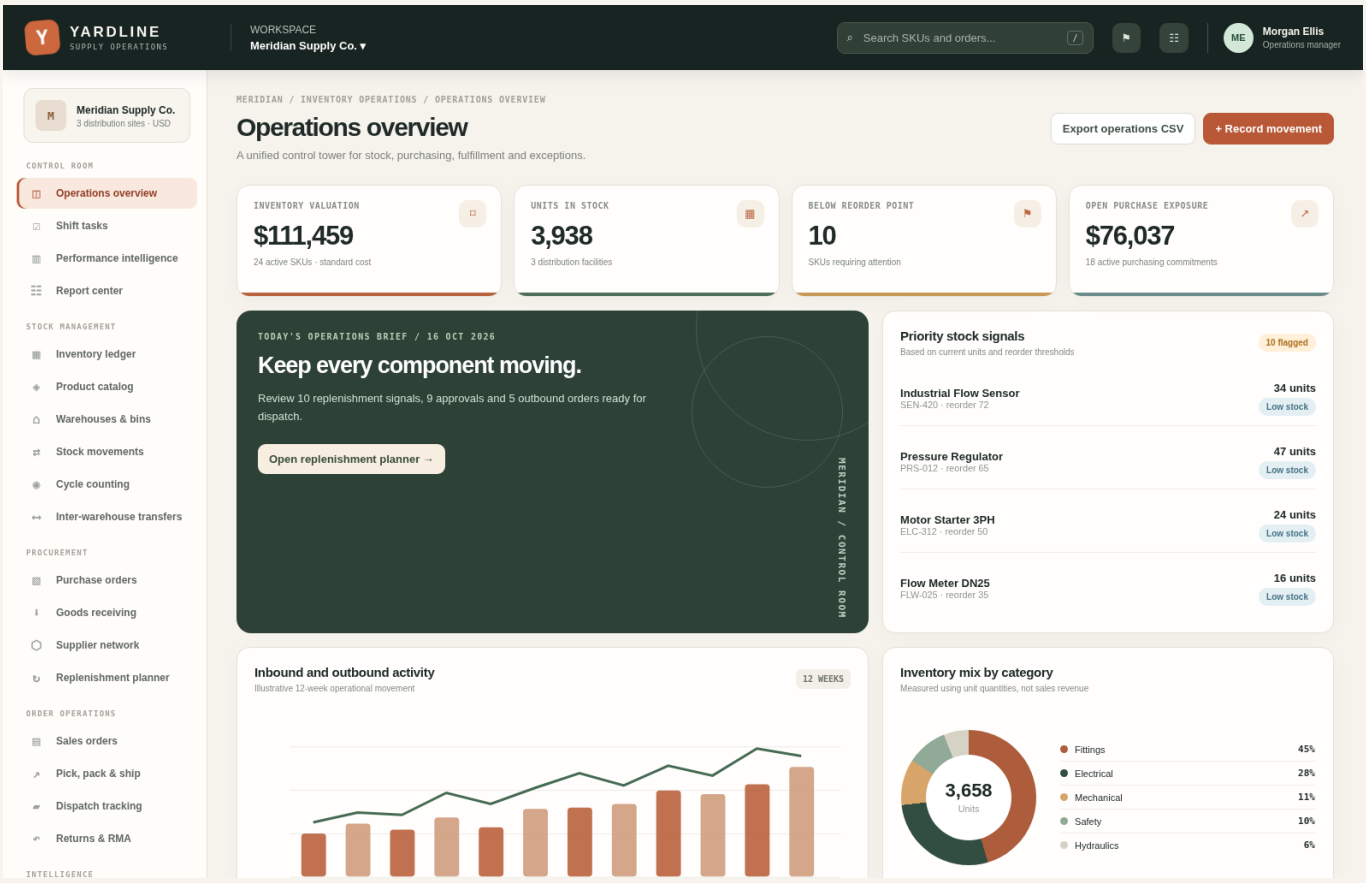
- Inventory ledger with 24 SKUs across 3 sample warehouse locations
- Goods receiving and transfer journal updates connected to warehouse balances
- Purchasing, order fulfillment, returns and cycle-count reconciliation
- Rule-based replenishment planner with draft purchase-order creation
- Approval desk, stock exceptions, analytics, report downloads and audit views
- n8n starter workflow to demonstrate safe purchase-request decisions

DESIGN LANGUAGE

Industrial operations control room: warm stone, deep forest, rust highlights, mono numeric references, structured tables, exception cues and compact operational cards.

02 / OPERATIONS OVERVIEW

At-a-glance financial valuation, current SKU stock, supplier commitments, low-stock signals and operational activity. KPIs are computed from the included sample records.

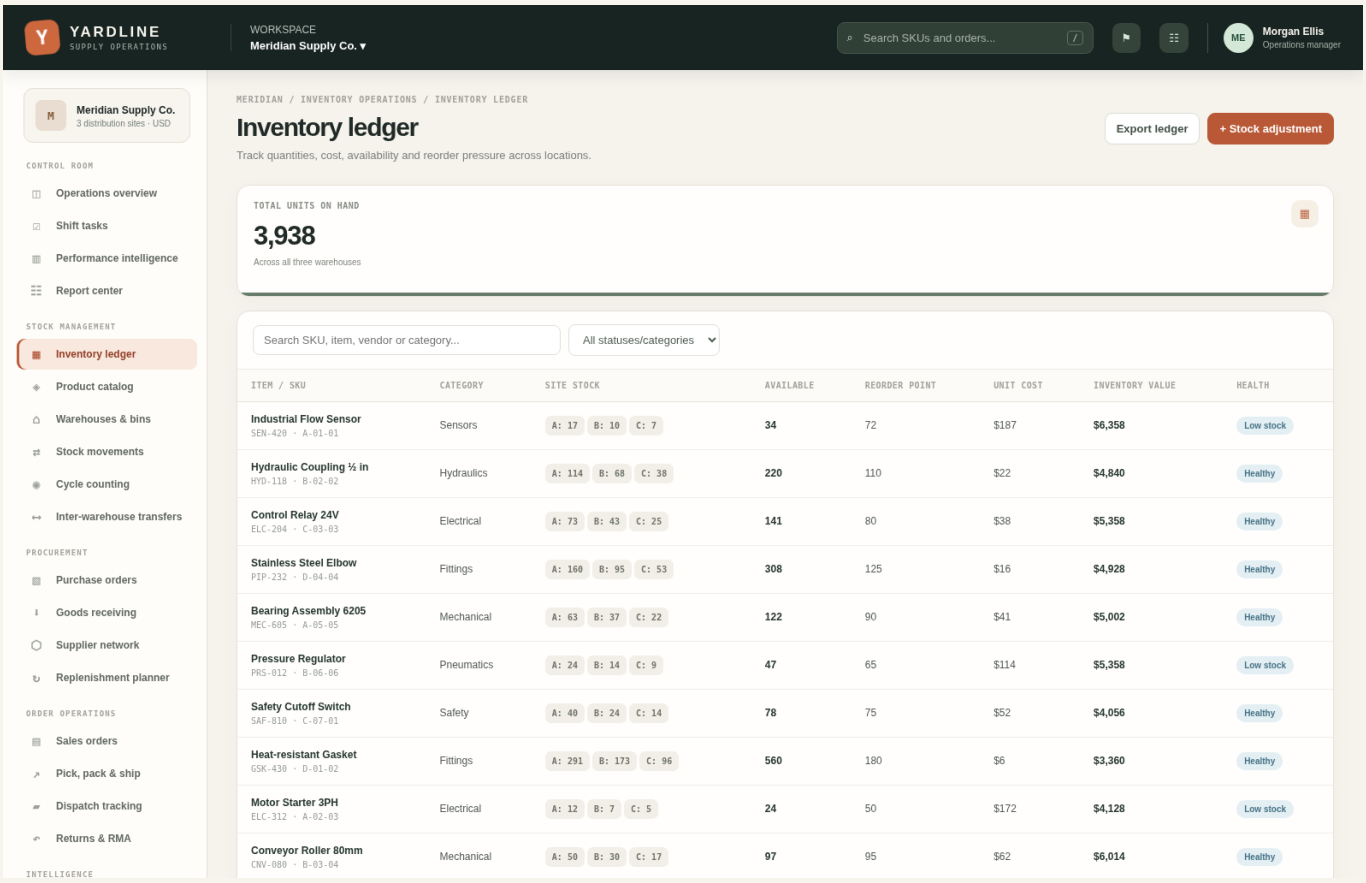


FUNCTIONAL HIGHLIGHTS

Calculated KPIs • Exception signals • Procurement activity

03 / INVENTORY LEDGER

Warehouse-specific stock quantities, reorder points, inventory valuation and product health with search and record inspection.

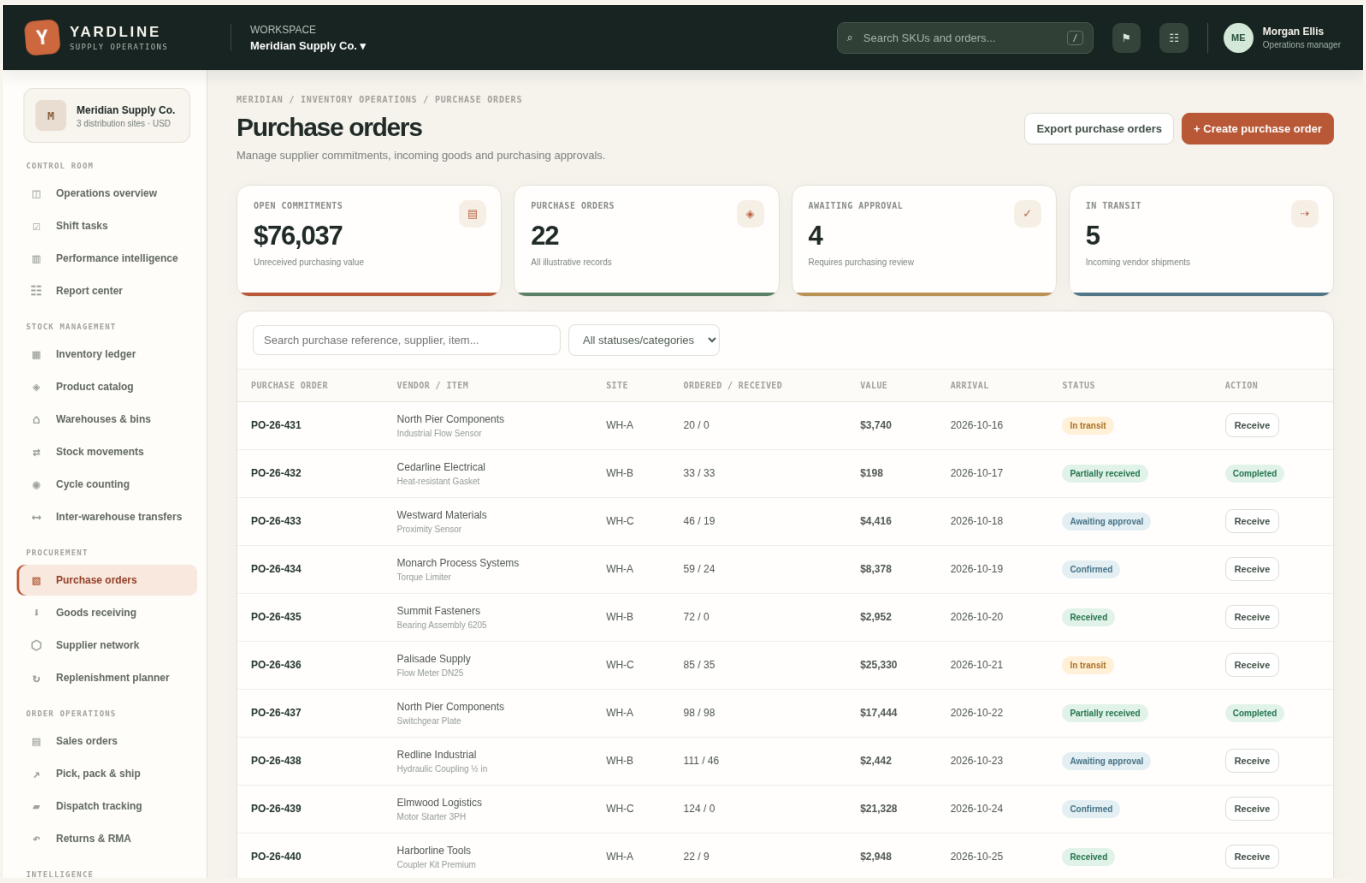


FUNCTIONAL HIGHLIGHTS

SKU search • Multi-warehouse quantities • Record inspection

04 / PURCHASING AND RECEIVING

Open purchase orders, outstanding cost exposure, expected arrivals, purchasing approval states and controlled stock-receipt actions.

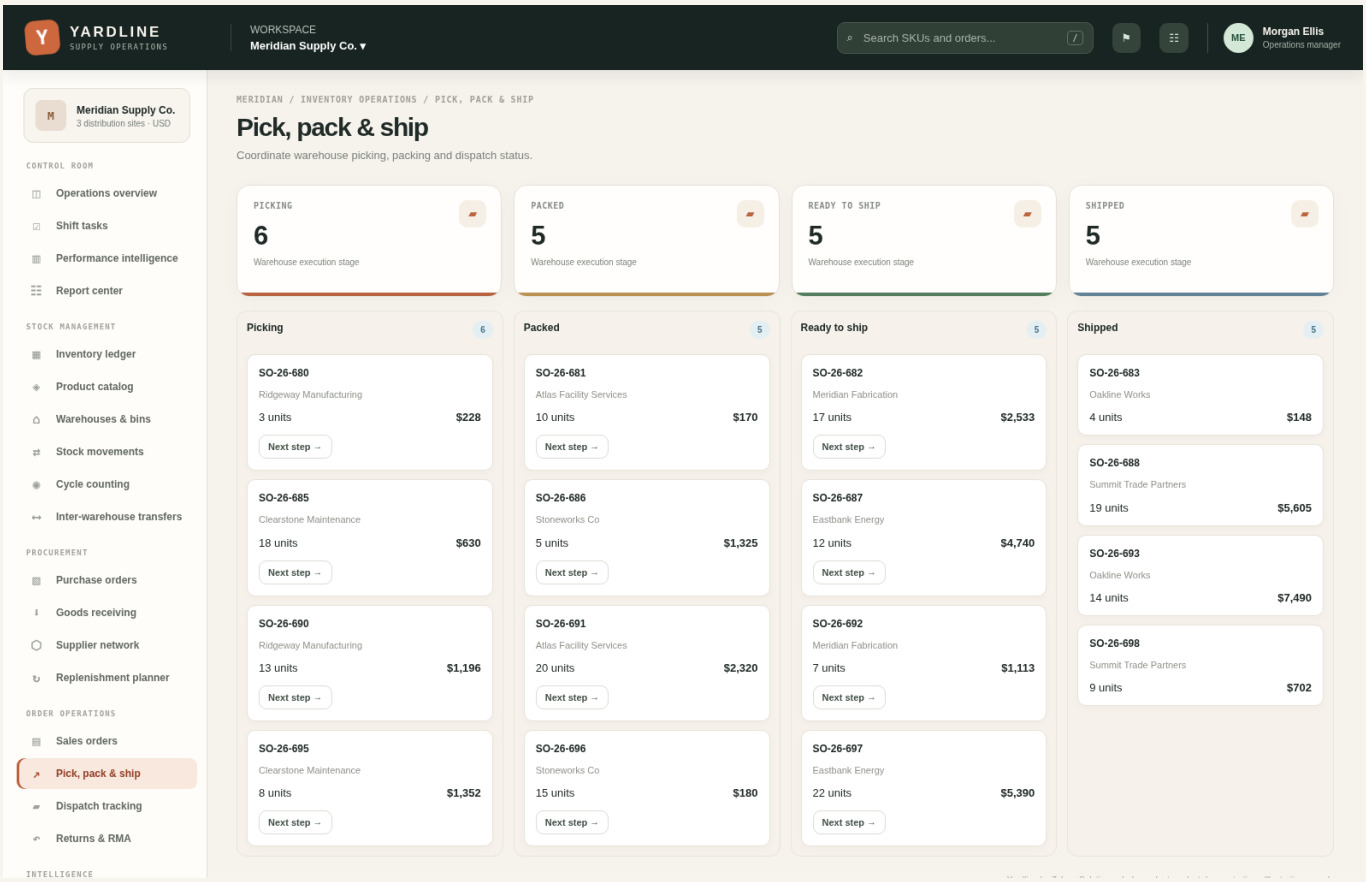


FUNCTIONAL HIGHLIGHTS

PO creation • Controlled receipts • Movement journal

05 / WAREHOUSE FULFILLMENT

Kanban order status, distribution-site information and forward-only order progress. Dispatch checks stock before posting an outbound movement.

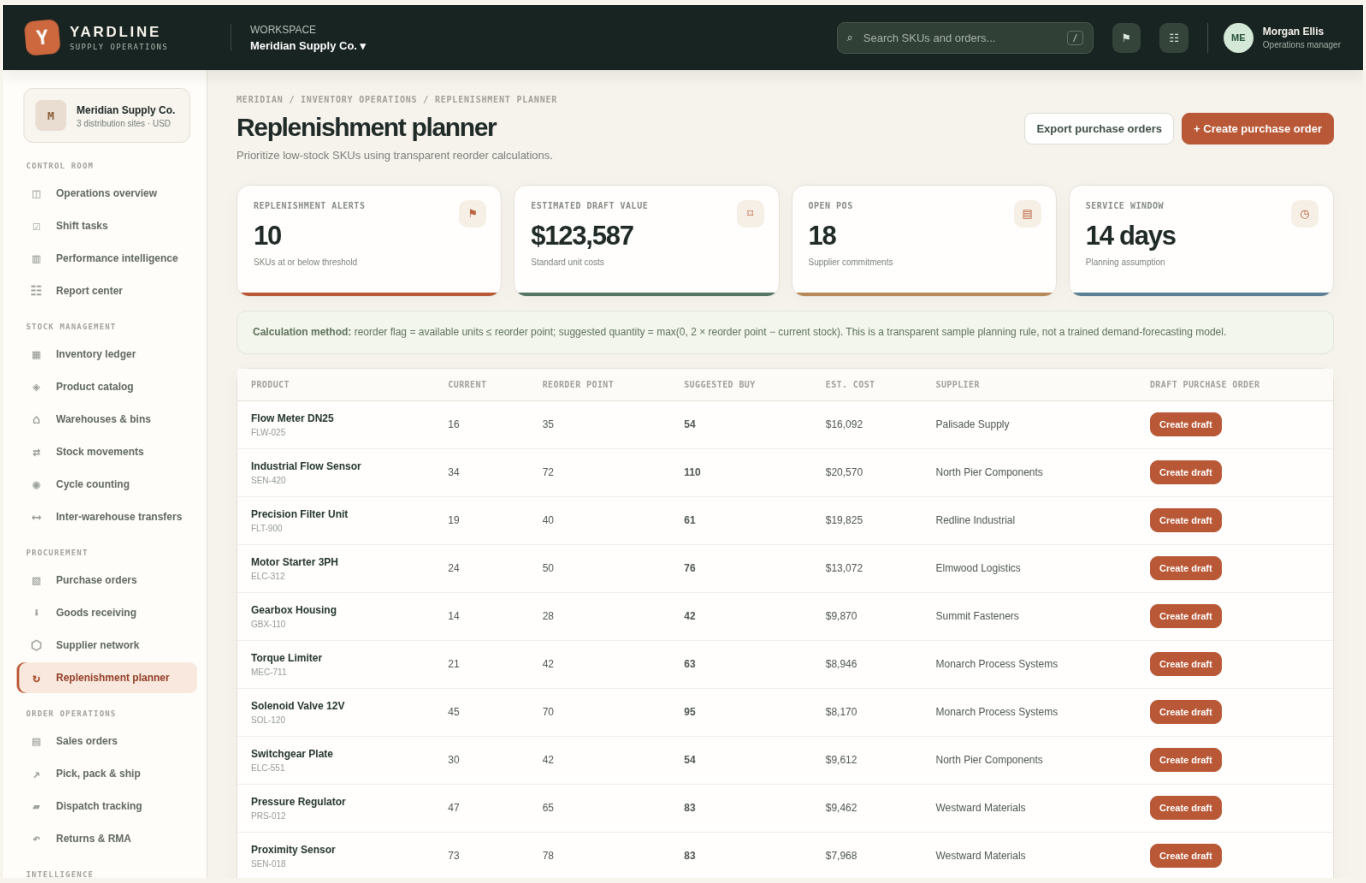


FUNCTIONAL HIGHLIGHTS

Order status transitions • Dispatch validation • Warehouse sourcing

06 / REPLENISHMENT INTELLIGENCE

Deterministic suggested purchases use current on-hand quantities and reorder thresholds. Draft purchase requests require review before any external step.

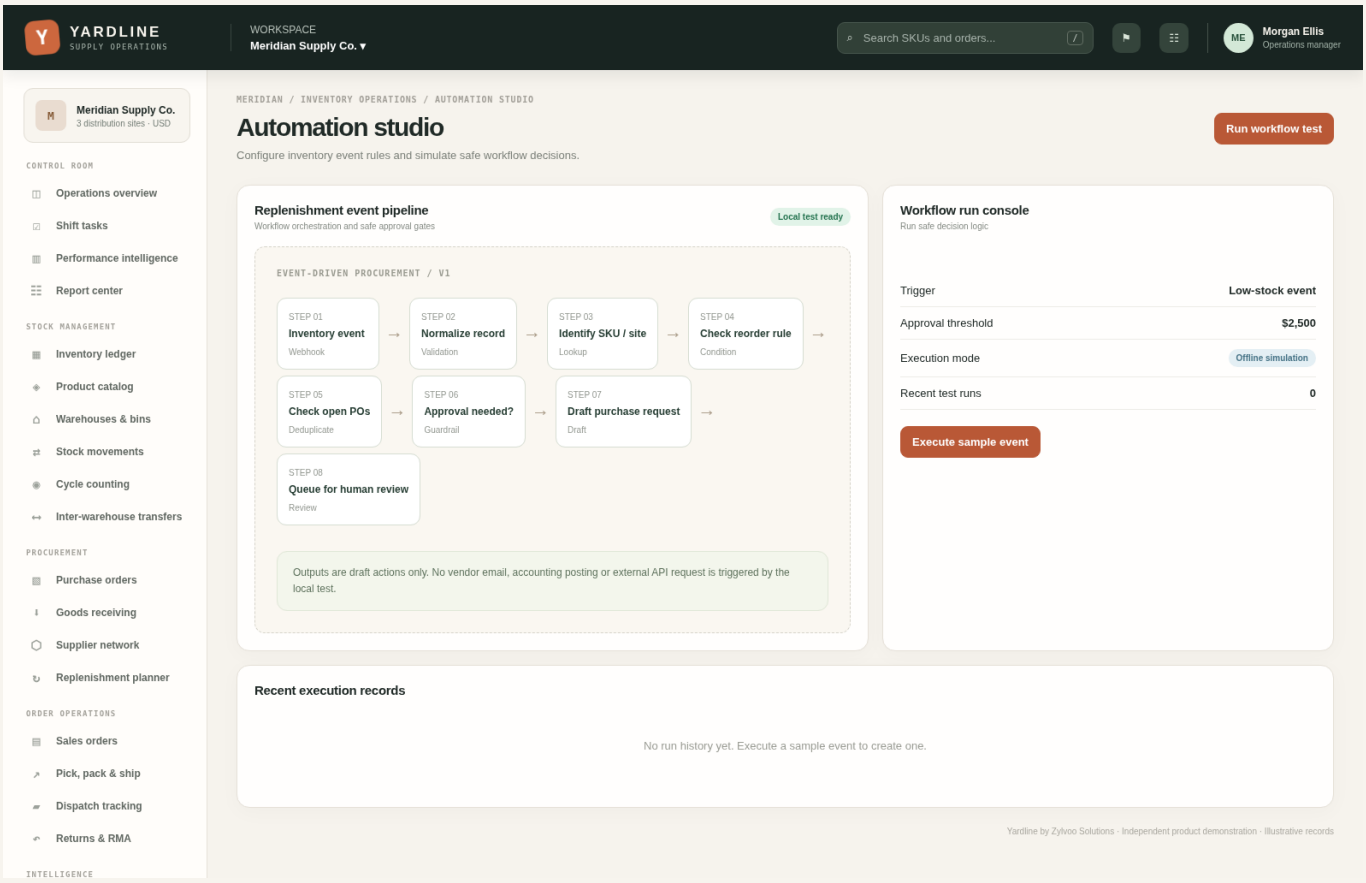


FUNCTIONAL HIGHLIGHTS

Rule-based decisions • Suggested buy quantities • Draft POs

07 / AUTOMATION STUDIO

Event-processing structure, validation gates, reorder calculations, workflow test logs and explicit human-review handoffs.

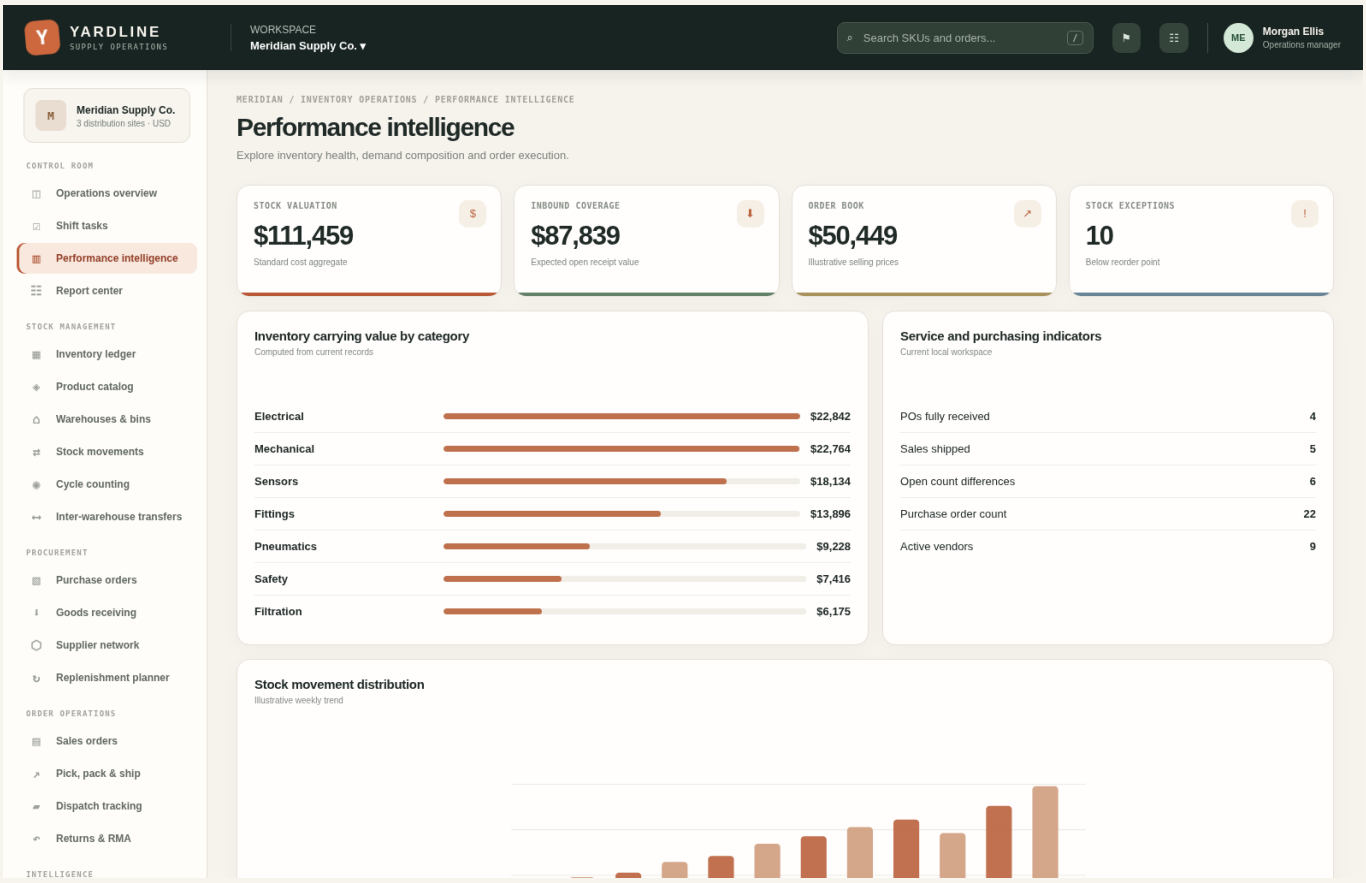


FUNCTIONAL HIGHLIGHTS

Validated decision logic • Test runs • Human-review routing

08 / INSIGHTS AND VALUATION

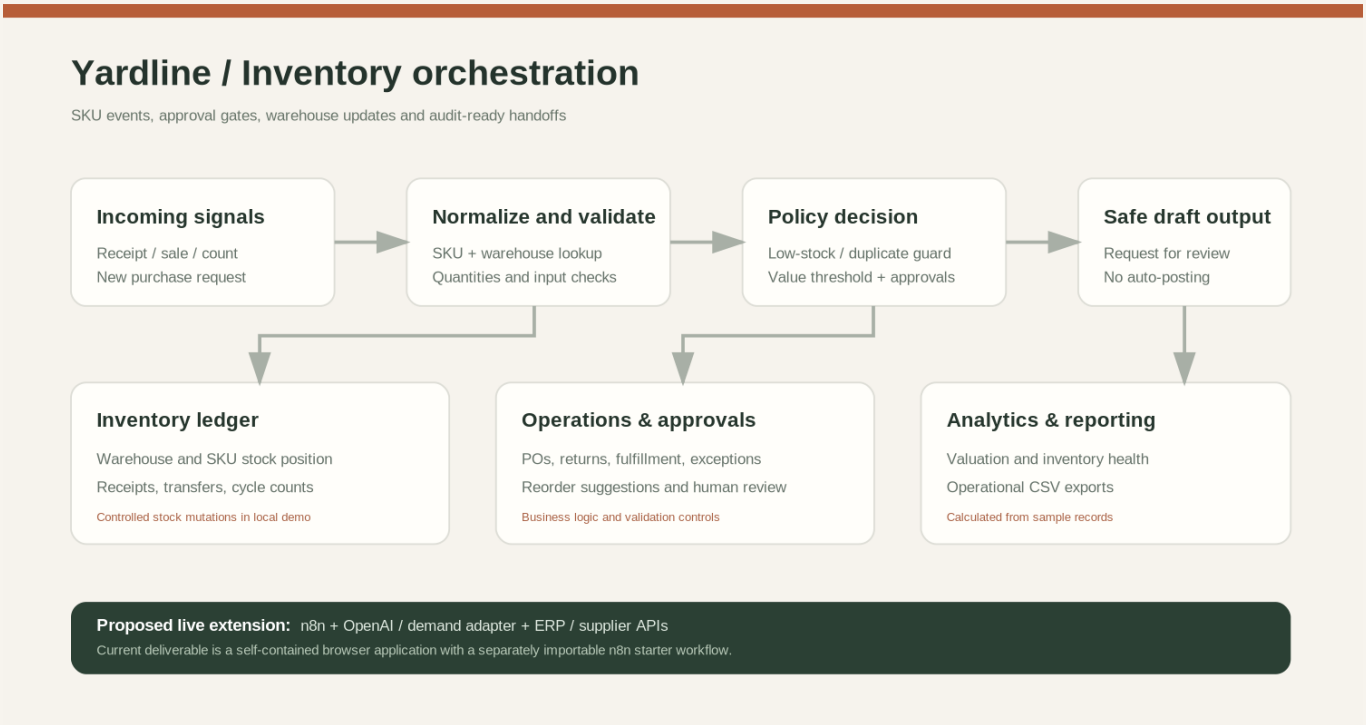
Management reporting with valuation, inbound purchase exposure, current order book, cost categories and operational KPIs.



FUNCTIONAL HIGHLIGHTS

Category analysis • Stock cost view • CSV reporting

09 / ARCHITECTURE & WORKFLOW



N8N STARTER DESIGN

Webhook inventory event → input validation → stock threshold decision → draft review payload → structured webhook response. There is no automatic purchase-order placement, supplier email or accounting posting.

Implementation controls before production:

- Authenticating source events, rate limiting and idempotent processing
- Maintaining real transactional stock balances in a database
- Adding vendor/ERP connectors, error queues, retry policy and monitoring
- Protecting customer records, managing roles and retaining audit logs